

SQL Queries and Results (PostgreSQL 18) for AI Job Impact Analytics Project

```
CREATE TABLE ai_on_job
(
    Employee_ID VARCHAR (10),
    Age INT,
    Gender VARCHAR (10),
    Education_Level VARCHAR (20),
    Industry VARCHAR (20),
    Job_Role VARCHAR (30),
    Years_Experience INT,
    AI_Adoption_Level VARCHAR (10),
    Automation_Risk VARCHAR (10),
    Upskilling_Required VARCHAR (5),
    Salary_Before_AI INT,
    Salary_After_AI INT,
    Job_Status VARCHAR (15),
    Work_Hours_Per_Week INT,
    Remote_Work VARCHAR (5),
    Job_Satisfaction INT,
    Productivity_Change NUMERIC (10,2)
);
```

```
COPY ai_on_job
FROM 'D:\New Datasets\ai_job_impact.csv'
DELIMITER ','
CSV HEADER;
```

```
SELECT * FROM ai_on_job;
```

	employee_id character varying (10)	age integer	gender character varying (10)	education_level character varying (20)	industry character varying (20)	job_role character varying (30)	years_experience integer	ai_adoption_level character varying (10)	automation_risk character varying (10)	upskilling_required character varying (5)	salary_before_ai integer	salary_a integer
1	E0001	50	Female	Bachelor	Marketing	Content Creator	25	High	High	Yes	106820	
2	E0002	45	Male	High School	Manufacturing	Quality Inspector	19	Low	Low	Yes	74131	
3	E0003	51	Female	Master	IT	DevOps Engineer	28	Medium	Medium	Yes	35311	
4	E0004	48	Male	PhD	Education	Teacher	24	Medium	Medium	Yes	114478	
5	E0005	24	Male	Bachelor	Healthcare	Doctor	0	High	Medium	No	33890	
6	E0006	46	Male	High School	Healthcare	Technician	21	Low	Medium	No	103969	
7	E0007	49	Male	PhD	Education	Administrator	24	High	Medium	No	78555	
8	E0008	57	Male	Bachelor	Finance	Auditor	34	Low	High	No	99479	
9	E0009	50	Other	Bachelor	Education	Professor	27	Low	High	Yes	102409	
10	E0010	52	Male	High School	Healthcare	Nurse	27	High	Low	Yes	31016	
11	E0011	45	Female	High School	Retail	Store Manager	22	Medium	High	No	93704	
12	E0012	36	Other	Master	Healthcare	Nurse	10	High	Low	Yes	78984	
13	E0013	32	Other	Bachelor	Healthcare	Nurse	8	High	High	No	69504	

```
-- * Average Salary Before AI : -
SELECT
    ROUND(AVG(Salary_Before_AI),2) AS Average_Salary_Before_AI_Adoption
FROM ai_on_job;
```

	average_salary_before_ai_adoption numeric	
1	73942.07	

```
-- * Average Salary After AI : -
SELECT
    ROUND(AVG(Salary_After_AI),2) AS Average_Salary_After_AI_Adoption
FROM ai_on_job;
```

	average_salary_after_ai_adoption
1	78428.64

```
-- * Change of Average Salary After AI Adoption : -
SELECT
    ROUND(AVG(Salary_Before_AI),2) AS Average_Salary_Before_AI_Adoption,
    ROUND(AVG(Salary_After_AI),2) AS Average_Salary_After_AI_Adoption,|
    ROUND(AVG(Salary_After_AI-Salary_Before_AI),2) AS Average_Change_of_Salary_After_AI_Adoption
FROM ai_on_job;
```

	average_salary_before_ai_adoption	average_salary_after_ai_adoption	average_change_of_salary_after_ai_adoption
1	73942.07	78428.64	4486.57

```
-- * Industries with Highest AI Adoption : -
SELECT
    Industry,
    AI_Adoption_Level,
    COUNT (*) AS Total_no_of_Employees
FROM ai_on_job
GROUP BY 1,2
ORDER BY Total_no_of_Employees DESC;
```

	industry	ai_adoption_level	total_no_of_employees
	character varying (20)	character varying (10)	bigint
1	Healthcare	High	113
2	Manufacturing	Medium	110
3	Education	Medium	107
4	Education	High	106
5	IT	Low	106
6	Finance	High	104
7	Retail	High	100
8	Finance	Low	100
9	IT	Medium	98
10	Marketing	High	97
11	Finance	Medium	96

```
-- * Industries with Highest Automation Risk : -
SELECT
    Industry,
    Automation_Risk,
    COUNT (*) AS Total_n0_of_Employees
FROM ai_on_job
WHERE Automation_Risk = 'High'
GROUP BY 1,2
ORDER BY 3 DESC;
```

	industry character varying (20) 🔒	automation_risk character varying (10) 🔒	total_n0_of_employees bigint 🔒
1	Education	High	112
2	Finance	High	103
3	Marketing	High	101
4	Manufacturing	High	98
5	IT	High	96
6	Retail	High	92
7	Healthcare	High	80

```
-- No. of Replaced Job Roles : -
SELECT
    Job_Role,
    Gender,
    COUNT (*) AS No_of_Replaced_Employees
FROM ai_on_job
WHERE Job_Status = 'Replaced'
GROUP BY Job_Role, Gender
ORDER BY No_of_Replaced_Employees DESC;
```

	job_role character varying (30) 🔒	gender character varying (10) 🔒	no_of_replaced_employees bigint 🔒
1	Accountant	Female	5
2	Teacher	Other	5
3	Machine Operator	Male	4
4	SEO Specialist	Male	4
5	Store Manager	Female	4
6	Investment Analyst	Female	4
7	Supervisor	Male	3
8	Digital Marketer	Female	3
9	Investment Analyst	Male	3
10	Content Creator	Other	3
11	Administrator	Other	3

```
-- * No. of Modified Job Roles : -

SELECT
    Job_Role,
    Gender,
    COUNT (*) AS No_of_Modified_Employees
FROM ai_on_job
WHERE Job_Status = 'Modified'
GROUP BY Job_Role, Gender
ORDER BY No_of_Modified_Employees DESC;
```

	job_role character varying (30) 🔒	gender character varying (10) 🔒	no_of_modified_employees bigint 🔒
1	Technician	Male	25
2	Data Analyst	Other	20
3	Accountant	Other	20
4	Software Engineer	Other	18
5	Accountant	Male	18
6	Store Manager	Other	18
7	Auditor	Male	18
8	Auditor	Female	18
9	DevOps Engineer	Other	17
10	SEO Specialist	Male	17
11	Teacher	Other	17

```
-- * No. of Unchanged Job Roles : -

SELECT
    Job_Role,
    Gender,
    COUNT (*) AS No_of_Unchanged_Employees
FROM ai_on_job
WHERE Job_Status = 'Unchanged'
GROUP BY Job_Role, Gender
ORDER BY No_of_Unchanged_Employees DESC;
```

	job_role character varying (30) 🔒	gender character varying (10) 🔒	no_of_unchanged_employees bigint 🔒
1	Professor	Male	34
2	Investment Analyst	Other	28
3	Supervisor	Male	26
4	DevOps Engineer	Female	26
5	Machine Operator	Male	25
6	Quality Inspector	Male	23
7	Administrator	Male	23
8	Quality Inspector	Other	23
9	Store Manager	Other	22
10	Quality Inspector	Female	21
11	Content Creator	Female	21

```
-- * Impact of AI on Productivity : -
SELECT
    AI_Adoption_Level,
    ROUND (AVG(Productivity_Change),2) AS Avg_Productivity_Change
FROM ai_on_job
GROUP BY 1
ORDER BY 2;
```

	ai_adoption_level character varying (10) 🔒	avg_productivity_change numeric 🔒
1	Low	8.03
2	Medium	10.17
3	High	10.95

```
-- * No. of Employees with High AI Adoption Level : -
SELECT
    Gender,
    Age,
    AI_Adoption_Level,
    COUNT (*) AS Total_no_Of_Employees
FROM ai_on_job
WHERE AI_Adoption_Level = 'High'
GROUP BY 1,2,3
ORDER BY 2 DESC;
```

	gender character varying (10) 🔒	age integer 🔒	ai_adoption_level character varying (10) 🔒	total_no_of_employees bigint 🔒
1	Other	59	High	5
2	Female	59	High	8
3	Male	59	High	7
4	Male	58	High	9
5	Other	58	High	3
6	Female	58	High	5
7	Other	57	High	9
8	Female	57	High	6
9	Male	57	High	8
10	Other	56	High	4
11	Female	56	High	5

```
-- * Salary Growth After AI Adoption : -
SELECT
    Gender,
    Age,
    ROUND(AVG(Salary_After_AI-Salary_Before_AI),2) AS Average_Change_of_Salary_After_AI_Adoption,
    COUNT (*) AS Total_no_of_Employees
FROM ai_on_job
GROUP BY Gender, Age
ORDER BY Average_Change_of_Salary_After_AI_Adoption DESC;
```

	gender character varying (10) 🔒	age integer 🔒	average_change_of_salary_after_ai_adoption numeric 🔒	total_no_of_employees bigint 🔒
1	Other	52	11952.71	17
2	Male	44	10464.06	17
3	Other	25	9534.63	19
4	Other	28	9345.71	7
5	Female	25	9255.00	19
6	Other	44	8622.80	20
7	Male	35	8315.95	19
8	Male	30	8177.64	11
9	Male	52	7876.62	13
10	Other	32	7732.41	17
11	Female	29	7653.27	22

```
-- * Productivity of Remote Workers : -
SELECT
    Remote_Work,
    Job_Role,
    ROUND (AVG(Productivity_Change),2) AS Avg_Productivity_Change,
    COUNT (*) AS Total_no_of_Employees
FROM ai_on_job
WHERE Remote_Work = 'Yes'
GROUP BY 1,2
ORDER BY 3 DESC;
```

	remote_work character varying (5) 🔒	job_role character varying (30) 🔒	avg_productivity_change numeric 🔒	total_no_of_employees bigint 🔒
1	Yes	Quality Inspector	12.20	64
2	Yes	Machine Operator	11.98	47
3	Yes	Content Creator	11.95	40
4	Yes	Professor	11.77	48
5	Yes	Doctor	11.65	39
6	Yes	DevOps Engineer	10.88	45
7	Yes	SEO Specialist	10.67	44
8	Yes	Administrator	10.58	50
9	Yes	Auditor	10.36	51
10	Yes	Digital Marketer	10.34	43
11	Yes	Data Analyst	10.32	48

```
-- * Remote Work Trends : -
SELECT
Industry,
COUNT (*) AS No_of_Remote_Workers
FROM ai_on_job
WHERE Remote_Work = 'Yes'
GROUP BY Industry
ORDER BY No_of_Remote_Workers;
```

	industry character varying (20) 🔒	no_of_remote_workers bigint 🔒
1	Healthcare	125
2	Marketing	127
3	Retail	134
4	IT	137
5	Education	142
6	Finance	150
7	Manufacturing	158

```
-- * Industries Requiring Upskilling of Employees : -
SELECT
    Industry,
    Job_Role,
    COUNT (*) AS No_of_Employees_needing_Upskilling
FROM ai_on_job
WHERE Upskilling_Required = 'Yes'
GROUP BY 1,2
ORDER BY 3;
```

	industry character varying (20) 🔒	job_role character varying (30) 🔒	no_of_employees_needing_upskilling bigint 🔒
1	Retail	Sales Associate	30
2	Retail	Cashier	36
3	Healthcare	Doctor	43
4	Healthcare	Technician	43
5	Education	Teacher	44
6	Manufacturing	Supervisor	45
7	Education	Professor	46
8	Marketing	SEO Specialist	48
9	IT	Software Engineer	49
10	Marketing	Content Creator	49
11	Finance	Accountant	50

```
-- * Average Job Satisfaction of Employees After AI Adoption : -
SELECT
    Job_Role,
    AI_Adoption_Level,
    ROUND(AVG(Job_Satisfaction),2) AS Avg_Job_Satisfaction
FROM ai_on_job
GROUP BY 1,2
ORDER BY 3 DESC;
```

	job_role character varying (30) 🔒	ai_adoption_level character varying (10) 🔒	avg_job_satisfaction numeric 🔒
1	DevOps Engineer	High	6.96
2	Machine Operator	Medium	6.72
3	Sales Associate	Medium	6.71
4	Digital Marketer	Medium	6.69
5	Sales Associate	Low	6.64
6	Professor	High	6.55
7	Doctor	Low	6.54
8	Machine Operator	Low	6.50
9	Software Engineer	High	6.43
10	Store Manager	Low	6.39
11	Doctor	Medium	6.38


```
-- * Top 10 Industries with Longest Average Working Hours Per Week : -
SELECT
    Industry,
    Job_Role,
    Remote_Work,
    ROUND (AVG(Work_Hours_Per_Week),2) AS Avg_Working_Hours_per_Week
FROM ai_on_job
GROUP BY 1,2,3
ORDER BY 4 DESC
LIMIT 10;
```

	industry character varying (20) 🔒	job_role character varying (30) 🔒	remote_work character varying (5) 🔒	avg_working_hours_per_week numeric 🔒
1	Finance	Accountant	No	46.92
2	Retail	Sales Associate	No	46.38
3	IT	Software Engineer	Yes	46.32
4	Healthcare	Technician	Yes	45.89
5	Finance	Auditor	No	45.73
6	Education	Teacher	No	45.70
7	Healthcare	Nurse	Yes	45.68
8	Education	Teacher	Yes	45.59
9	Marketing	Digital Marketer	Yes	45.56
10	Manufacturing	Machine Operator	Yes	45.55

```
-- * Experience Wise AI Adoption Level : -
SELECT
    CASE WHEN Years_Experience < 5 THEN '0 - 5 Years Experience'
    WHEN Years_Experience BETWEEN 5 AND 10 THEN '5 - 10 Years Experience'
    WHEN Years_Experience BETWEEN 11 AND 20 THEN '11 - 20 Years Experience'
    WHEN Years_Experience BETWEEN 21 AND 30 THEN '21 - 30 Years Experience'
    ELSE '30+ Years Experience' END AS Experience_Category,

    AI_Adoption_Level,
    Age,
    COUNT (*) AS Total_no_of_Employees
FROM ai_on_job
GROUP BY 1, 2, 3
ORDER BY 2,4 DESC;
```

	experience_category	ai_adoption_level	age	total_no_of_employees
	text	character varying (10)	integer	bigint
1	11 - 20 Years Experien...	High	38	29
2	21 - 30 Years Experien...	High	47	25
3	21 - 30 Years Experien...	High	49	24
4	11 - 20 Years Experien...	High	40	24
5	0 - 5 Years Experience	High	25	23
6	30+ Years Experience	High	57	23
7	21 - 30 Years Experien...	High	50	20
8	11 - 20 Years Experien...	High	37	20
9	30+ Years Experience	High	59	20
10	0 - 5 Years Experience	High	23	19
11	11 - 20 Years Experien...	High	43	19

```
-- * Employees with Decreased Salary After Adoption of AI : -
SELECT
    Employee_ID,
    Industry,
    Education_Level,
    Job_Role,
    Years_Experience,
    Salary_Before_AI,
    Salary_After_AI,
    (Salary_After_AI-Salary_Before_AI) AS Change_of_Salary
FROM ai_on_job
WHERE Salary_After_AI < Salary_Before_AI
ORDER BY Change_of_Salary DESC;
```

	employee_id	industry	education_level	job_role	years_experience	salary_before_ai	salary_after_ai	change_of_salary
	character varying (10)	character varying (20)	character varying (20)	character varying (30)	integer	integer	integer	integer
1	E1803	Finance	Master	Auditor	34	50128	50105	-23
2	E0626	IT	PhD	Software Engineer	33	41151	41119	-32
3	E1779	Manufacturing	High School	Quality Inspector	16	82948	82914	-34
4	E0079	Marketing	High School	SEO Specialist	25	85129	85089	-40
5	E1735	Marketing	Master	SEO Specialist	34	32466	32426	-40
6	E1045	Finance	PhD	Auditor	30	30453	30407	-46
7	E1929	Marketing	Bachelor	Content Creator	7	76221	76174	-47
8	E0806	Manufacturing	PhD	Quality Inspector	1	49581	49529	-52
9	E1579	Retail	High School	Sales Associate	34	94814	94757	-57
10	E0658	Manufacturing	Bachelor	Supervisor	1	40377	40297	-80
11	E1261	Manufacturing	Master	Quality Inspector	13	52391	52305	-86

```
-- * Most Common Job Status : -
SELECT
    Job_Status,
    Job_Role,
    Years_Experience,
    COUNT (*) AS Total_no_of_Employees
FROM ai_on_job
GROUP BY 1, 2, 3
ORDER BY Total_no_of_Employees DESC;
```

	job_status character varying (15) 🔒	job_role character varying (30) 🔒	years_experience integer 🔒	total_no_of_employees bigint 🔒
1	Unchanged	Digital Marketer	0	8
2	Unchanged	Machine Operator	0	8
3	Unchanged	Teacher	0	8
4	Unchanged	Sales Associate	0	7
5	Unchanged	Professor	0	7
6	Unchanged	Quality Inspector	0	6
7	Modified	Data Analyst	0	6
8	Unchanged	Supervisor	0	6
9	Modified	DevOps Engineer	0	6
10	Unchanged	Software Engineer	19	6
11	Unchanged	Investment Analyst	0	6

```
-- * Top 50 Employees with Highest Productivity : -
SELECT
    Employee_ID ,
    Age,
    Gender,
    Industry,
    Job_Role,
    Years_Experience,
    Productivity_Change
FROM ai_on_job
ORDER BY Productivity_Change DESC
LIMIT 50;
```

	employee_id character varying (10) 🔒	age integer 🔒	gender character varying (10) 🔒	industry character varying (20) 🔒	job_role character varying (30) 🔒	years_experience integer 🔒	productivity_change numeric (10,2) 🔒
1	E1174	42	Other	IT	Software Engineer	20	39.99
2	E0636	23	Other	Manufacturing	Machine Operator	0	39.97
3	E1993	31	Male	Marketing	Content Creator	7	39.91
4	E0982	37	Male	Finance	Investment Analyst	12	39.90
5	E1036	53	Male	Manufacturing	Quality Inspector	29	39.88
6	E0024	23	Female	Retail	Store Manager	0	39.86
7	E1836	56	Female	Education	Teacher	33	39.83
8	E0794	39	Male	Manufacturing	Quality Inspector	16	39.82
9	E1569	55	Female	Healthcare	Doctor	30	39.82
10	E0913	43	Male	Education	Professor	21	39.81
11	E1340	37	Male	Manufacturing	Quality Inspector	12	39.80

```
-- * Productivity and Job Status : -
SELECT
    Job_Status,
    ROUND (AVG(Productivity_Change),2) AS Avg_Productivity_Change,
    COUNT (*) Total_no_of_Employees
FROM ai_on_job
GROUP BY 1
ORDER BY 2, 3 DESC;
```

	job_status character varying (15) 🔒	avg_productivity_change numeric 🔒	total_no_of_employees bigint 🔒
1	Unchanged	9.47	1093
2	Modified	10.12	801
3	Replaced	10.56	106

```
-- * Rank of Industries On The Basis of Average Salary After Adoption of AI : -
SELECT
    Industry,
    ROUND(AVG(Salary_After_AI),2) AS Average_Salary_After_AI_Adoption,
    RANK () OVER (ORDER BY ROUND(AVG(Salary_After_AI),2) DESC) AS Rank_of_Salary
FROM ai_on_job
GROUP BY Industry;
```

	industry character varying (20) 🔒	average_salary_after_ai_adoption numeric 🔒	rank_of_salary bigint 🔒
1	Retail	80376.96	1
2	IT	79551.60	2
3	Healthcare	78702.03	3
4	Manufacturing	78489.94	4
5	Education	78322.74	5
6	Marketing	77597.73	6
7	Finance	76106.43	7

```
-- * Top 5 Highest Paid Jobs In Each Industry : -
SELECT
*
FROM
(
SELECT
Job_Role,
Job_status,
Industry,
ROUND(AVG(Salary_After_AI),2) AS Average_Salary_After_AI_Adoption,
RANK () OVER(PARTITION BY Industry ORDER BY ROUND(AVG(Salary_After_AI),2) DESC) AS Rank_of_Salary
FROM ai_on_job
GROUP BY 1, 2, 3
) AS Rank_of_Jobs
WHERE Rank_of_Salary <= 5;
```

	job_role character varying (30) 🔒	job_status character varying (15) 🔒	industry character varying (20) 🔒	average_salary_after_ai_adoption numeric 🔒	rank_of_salary bigint 🔒
1	Administrator	Replaced	Education	95755.88	1
2	Teacher	Modified	Education	85044.54	2
3	Administrator	Modified	Education	79312.07	3
4	Administrator	Unchanged	Education	78123.88	4
5	Teacher	Replaced	Education	76563.00	5
6	Investment Analyst	Replaced	Finance	108487.25	1
7	Accountant	Replaced	Finance	78267.00	2
8	Investment Analyst	Unchanged	Finance	77662.65	3
9	Accountant	Modified	Finance	77333.96	4
10	Accountant	Unchanged	Finance	76016.77	5
11	Technician	Replaced	Healthcare	93959.00	1

```
-- * Percentage of Job Replacement by AI : -
SELECT
ROUND (100 * COUNT (*) FILTER (WHERE Job_status = 'Replaced') / COUNT (*) , 2) AS Percentage_of_Job_Replacement
FROM ai_on_job;
```

	percentage_of_job_replacement numeric 🔒
1	5.00